

MTH 553: Lecture # 36

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Lecture Span

1. Dirichlet's Principle

Here, we attempt to solve Poisson's equations by minimizing energy. Let Ω be bounded and smooth in $\mathbb{R}^n$. Let $f \in C(\bar{\Omega})$ and $g \in C(\partial\Omega)$.

Define

$$E[w] = \int_{\Omega} \left(\frac{1}{2} |\nabla w|^2 - wf \right) dx \quad (1)$$

For w in the set

$$A = \{w \in C^2(\bar{\Omega}) : w = g \text{ on } \partial\Omega\} \quad (2)$$

Here, we assume that $A \neq \emptyset$. Note that w is really any arbitrary function that equals g on $\partial\Omega$.

Theorem 0.1. *If*

$$-\Delta u = f, \quad \Omega \quad (3)$$

$$u = g, \quad \partial\Omega \quad (4)$$

Then, Dirichlet's principle states that:

$$E[u] = \min_{w \in A} E[w] \quad (5)$$

Note that if $u \in A$ and also satisfies Dirichlet's Principle, then

$$-\Delta u = f, \quad \Omega \quad (6)$$

$$u = g, \quad \partial\Omega \quad (7)$$

Therefore,

$$\text{DP} \iff u \text{ solves Poisson's equations}$$

Proof. (a) Let $-\Delta u = f$. Then for all $w \in A$, then we have

$$0 = \int_{\Omega} (-\Delta u - f)(u - w) dx \quad (8)$$

$$= \int_{\Omega} [\nabla u \cdot \nabla(u - w) - f(u - w)] \quad (9)$$

By green 1 since $u = g = w$ on $\partial\Omega$. Then

$$= \int_{\Omega} [|\nabla u|^2 - \nabla u \cdot \nabla w - fu + fw] dx \quad (10)$$

$$\geq \int_{\Omega} \left[\frac{1}{2} |\nabla u|^2 - \frac{1}{2} |\nabla w|^2 - fu + fw \right] dx \quad (11)$$

Therefore,

$$E[w] \geq E[u] \quad (12)$$

Which implies the Dirichlet principle. $\square$

We next prove $\Leftarrow$.

Proof. (b) Fix $v \in C_0^\infty(\Omega)$ and let

$$e(\tau) = E[u + \tau v], \quad \tau \in \mathbb{R} \quad (13)$$

Since $u + \tau v \in A$, the DP implies

$$e(0) \leq e(\tau) \implies e'(0) = 0 \quad (14)$$

Then

$$0 = \frac{d}{dt} \int_{\Omega} \left[\frac{1}{2} |\nabla u|^2 + \tau \nabla u \cdot \nabla v + \frac{\tau^2}{2} |\nabla v|^2 - u f - \tau v f \right] \quad (15)$$

$$= \int_{\Omega} [\nabla u \cdot \nabla v - v f] \quad (16)$$

$$= \int_{\Omega} (-\Delta u - f) v dx \quad (17)$$

By green 1 since $v = 0$ on $\partial\Omega$. Therefore, for all $v \in C_0^\infty(\Omega)$, we have that $-\Delta u - f = 0$ and $u \in A$, therefore, $u = g$ on $\partial\Omega$. $\square$